

Debojjal Bagchi

Doctoral Student and Graduate Research Assistant
Maseeh Department of Civil, Architectural, and Environmental Engineering
The University of Texas at Austin
Ernest Cockrell Jr. Hall (ECJ) 6.202
301 E. Dean Keeton St.
Austin, TX, 78712-1172

Cell: 737-304-9684
Email: debojjalb@utexas.edu
Web: debojjalb.github.io/

EDUCATION

Doctor of Philosophy

2023–2027

The University of Texas at Austin
Major: Civil Engineering (Transportation Systems)
GPA: 4.00 / 4.00
Advisor: [Dr. Stephen D. Boyles](#)

Master of Science in Engineering (Thesis)

2023–2025

The University of Texas at Austin
Major: Civil Engineering (Transportation Systems), *Minor:* Operations Research and Industrial Engineering
GPA: 4.00 / 4.00
Thesis: Error bounds for stochastic user equilibrium traffic assignment. [[Link](#)]
Winner: CUTC Milton Pikarsky Memorial Award for outstanding MS thesis in science and technology
Advisor: [Dr. Stephen D. Boyles](#); *Reader:* [Dr. Zhaomiao Guo](#)

Bachelor of Science (Research)

2019–2023

Indian Institute of Science, Bengaluru
Major: Earth & Environmental Science, *Minor:* Mathematics
GPA: 8.7 / 10 (*Major GPA:* 9.4 / 10.0)
Thesis: Energy efficient and safe routing for last mile logistics. [[Link](#)]
Winner: Institute Gold Medal for highest major GPA
Advisor: [Dr. Tarun Rambha](#)

PROFESSIONAL SERVICE

- **Committee Member**, Transportation Research Board (TRB) Standing Committee AT014 on Intermodal Freight and Truck Transportation (2025–2028).
- **Committee Member**, Transportation Research Board (TRB) Standing Committee AW010 on Ports and Channels (2025, prior to restructuring).

AWARDS AND FELLOWSHIPS

International & National Awards

- **Milton Pikarsky Memorial Award**, Outstanding MS Thesis in Science and Technology, Council of University Transportation Centers, USA (2025).
- **MITACS Globalink Research Internship (GRI) Award**, Fully Funded Summer Research at a Canadian University, MITACS, Canada (2022).
- **Dhirubhai Ambani Scholarship**, Reliance Foundation, India (2019).
- **Kishore Vaigyanik Protsahan Yojana (KVPY) Fellowship**, Department of Science and Technology, Government of India (2019).
- **National Talent Search Examination (NTSE) Scholarship**, National Council of Educational Research and Training, India (2017).

Regional & University-Level Awards

- **Second Prize**, TexITE Student Paper Award, UT Austin (2026).
- **Best Poster Award**, Graduate Symposium, Department of Civil, Architectural and Environmental Engineering, UT Austin (2026).
- **Professional Development Award**, Graduate School, UT Austin (2025).
- **Graduate School Fellowship**, UT Austin (2023–2027).
- **Institute Gold Medal**, Indian Institute of Science, for highest major GPA in B.S. (Research).
- **Special Honor**, Academic Excellence by a Student, Telegraph School Awards (2019).
- **Jagadish Bose National Talent Search (JBNSTS) Fellowship** (2018).

ONGOING RESEARCH

Graduate Research Assistantship

Jan 2026 – Present

“Develop Multi-Modal Maritime-Rail-Roadway Transportation Model for the Texas Inland and Intercoastal Waterways” sponsored by Texas Department of Transportation, *PI: Dr. Stephen Boyles*

Graduate Research Assistantship

Jan 2026 – Present

“Evaluating Texas Ports Readiness and Opportunities for Alternative Fuels” sponsored by Texas Department of Transportation, *PI: Dr. Andrea Gold*

PAST RESEARCH EXPERIENCE

Graduate Research Assistantship

Sept 2023 – Dec 2024

“Data-Driven Multimodal Freight Modeling for Waterways and Port” sponsored by Coastal and Hydraulics Laboratory, US Army Corps of Engineers Research and Development Center, *PI: Dr. Stephen Boyles*

- Processed and integrated multimodal transportation data from AIS vessel tracking, terminal operations, and landside logistics and conducted stakeholder interviews as part of a research team.
- Built an extensive discrete event simulation using open-source software SimPy in Python for the entire Port of Houston system, including anchorage, channel, terminals, landside trucks, rail, and pipelines.
- Developed a queueing model for the Port of Houston to define operating capacity, and an ordinary differential equation-based model to identify system bottlenecks.
- Simulated disruption scenarios such as hurricanes and intermittent fog to assess port performance under capacity constraints and infrastructure expansion.
- **The work resulted in five conference presentations and six publications in preparation (four methodological, two empirical).**

MITACS Globalink Research Internship

May 2022 – Aug 2022

“Integrating Waste and Resource Management: Data-Driven Optimization of Urban Mining Logistics” sponsored by MITACS at Université du Québec à Trois-Rivières, *PI: Dr. Amina Lamghari*

- Performed an extensive literature review of heuristics for Reverse Logistics (RL) network design problems including Tabu search, simulated annealing, and bee colony optimization.
- Developed a scenario-based Mixed Integer Linear Program (MILP) formulation for the RL network design problem under uncertainties for wood industries of Quebec.
- Developed an Adaptive Large Neighbourhood Search (ALNS) heuristic for the RL network design problem and introduced the concept of adaptive neighborhoods.
- Solved the MILP using CPLEX and implemented the ALNS heuristic in Python.

Undergraduate Research

July 2021 – July 2023

“A Local Search Heuristic for the Bi-criteria Steiner Traveling Salesman Problem with Time Windows (BSTSPTW)” at the Indian Institute of Science, Bengaluru, India, *PI: Dr. Tarun Rambha*

- Performed extensive literature review of existing heuristic algorithms for the Traveling Salesman Problem, including Pareto Local Searches, Lin-Kernighan Heuristic, and r-opt.

- Developed and implemented two brute-force exact methods for the Bi-Objective Steiner Traveling Salesman Problem with Time Windows (BSTSPTW).
- Formulated a new scalarization-based Mixed Integer Program (MIP) formulation for BSTSPTW using a single commodity flow based formulation and implemented it using CPLEX.
- Proposed a novel local search-based heuristic with six operators that out-performed an adapted version of Lin-Kernighan-Helsgaun (LKH) heuristic on real world Amazon delivery routes in Austin, US.
- **The work resulted in two conference presentations and a journal article.**

PUBLICATIONS

- [M6] **Bagchi, D.** and Boyles, S. D. (2026). Spectral analysis of the logit mapping and implications for stochastic user equilibrium algorithms. *Under Review*. <https://doi.org/10.48550/arXiv.2605.21843>
- [M5] Robbennolt, J., **Bagchi, D.**, and Boyles, S. D. (2026). Dynamic traffic assignment with uncertain demand: A localized queue spillback approach. *Under Review*.
- [M4] **Bagchi, D.[†]**, Bathgate, K.[†], Mitchell, K.N., Asborn, M.I., Kress, M.M., and Boyles, S.D. (2025). Measuring capacities in multimodal maritime port systems with anchorage queues. *Under Review*. <https://doi.org/10.48550/arXiv.2509.22961>
- [M3] Bathgate, K., **Bagchi, D.**, Mitchell, K.N., Asborn, M.I., Kress, M.M., and Boyles, S.D. (2026). Anchorage Queue Dynamics due to Fog Channel Closures at the Port of Houston. *Journal of Waterway, Port, Coastal, and Ocean Engineering*. <https://doi.org/10.1061/jwped5.wweng-2451>
- [M2] Bathgate, K., **Bagchi, D.**, Mitchell, K.N., Asborn, M.I., and Kress, M.M., and Boyles, S.D. (2026). Characterizing Longitudinal Vessel Arrival and Queue Behavior from AIS at the Port of Houston. *Journal of Waterway, Port, Coastal, and Ocean Engineering*. <https://doi.org/10.1061/jwped5.wweng-2441>
- [M1] Agarwal, P.[†], **Bagchi, D.[†]**, Rambha, T., and Pandey, V. (2025). A Bi-criterion Steiner Traveling Salesperson Problem with Time Windows for Last-Mile Electric Vehicle Logistics. *Computers & Operations Research*. <https://doi.org/10.1016/j.cor.2025.107286> [Code] ([†]Equal contribution)

WORKING PAPERS

- [W2] **Bagchi, D.[†]**, Bathgate, K.[†], Mitchell, K.N., Asborn, M.I., and Kress, M.M., Boyles, S.D. (2026). A generalizable simulation framework for evaluating the resilience and capacity of multimodal port operations. [Code] ([†]Equal contribution)
- [W1] **Bagchi, D.** and Boyles, S. D. (2026). Error bounds for stochastic user equilibrium traffic assignment.

CONFERENCE PRESENTATIONS

- [C15] **Bagchi, D.** and S. D. (2026). Theoretical and Algorithmic Advances in Logit-Based Stochastic User Equilibrium Traffic Assignment. *Institute for Operations Research and the Management Sciences Annual Meeting 2025 (INFORMS)*, San Francisco, USA. (Invited session)
- [C14] **Bagchi, D.**, Bathgate, K., Boyles, S. D., and Asborn, M. I. (2026). A dynamic ultimate capacity framework for seaport systems. *Transportation Research Board 104th Annual Meeting (TRB)*, Washington, D.C., USA. (Lectern session)
- [C13] Bathgate, K., **Bagchi, D.**, Boyles, S. D., Mitchell, K. N., Asborn, M. I., and Kress, M. M. (2026). Characterizing anchorage queue dynamics due to channel fog closures at the Port of Houston. *Transportation Research Board 104th Annual Meeting (TRB)*, Washington, D.C., USA. (Poster session)
- [C12] Robbennolt, J., **Bagchi, D.**, and Boyles, S. D. (2026). Dynamic traffic assignment with uncertain demand: A localized queue spillback approach. *Transportation Research Board 104th Annual Meeting (TRB)*, Washington, D.C., USA. (Poster session)

- [C11] Robbennolt, J, **Bagchi, D.** and Boyles, S. D. (2025). Localized Queue Spillback with Uncertain Demand. *10th International Symposium on Dynamic Traffic Assignment (DTA2025)*, Salerno, Italy.
- [C10] **Bagchi, D.** and Boyles, S. D. (2025). Error bounds for stochastic user equilibrium traffic assignment. *12th Triennial Symposium on Transportation Analysis conference (TRISTAN XII)*, Okinawa, Japan.
- [C9] **Bagchi, D.**, Bathgate, K., and Boyles, S. D. (2025). A framework for measuring maritime port system capacities with limited input data. *Institute for Operations Research and the Management Sciences Annual Meeting 2024 (INFORMS)*, Seattle, USA. **(Invited session)**
- [C8] Bathgate, K., **Bagchi, D.**, and Boyles S. D (2025). Simulation methods for measuring multimodal maritime freight system capacity, and resilience: A case study at the Port of Houston. *Presented in UT Austin Center for Transportation Research Annual Symposium, 2025*, Austin, USA.
- [C7] **Bagchi, D.**, Bathgate, K., and Boyles, S. D. (2025). A queuing-theory-based operating capacity model for multimodal port operations. *Transportation Research Board 104th Annual Meeting 2023 (TRB)*, Washington, D.C., USA. **(Lectern session)**
- [C6] Bathgate, K., **Bagchi, D.**, and Boyles, S. D. (2025). Use of AIS data to characterize vessel mix in Houston port operations for simulation. *Transportation Research Board 104th Annual Meeting 2023 (TRB)*, Washington, D.C., USA. **(Lectern session)**
- [C5] Bathgate, K., **Bagchi, D.**, and Boyles, S. D. (2024). Identifying capacities in a multimodal maritime freight network. *Institute for Operations Research and the Management Sciences Annual Meeting 2024 (INFORMS)*, Seattle, USA. **(Invited session)**
- [C4] **Bagchi, D.**, and Boyles, S. D. (2024). Error Bounds for Stochastic User Equilibrium Traffic Assignment. *Institute for Operations Research and the Management Sciences Annual Meeting 2024 (INFORMS)*, Seattle, USA. **(Invited session)**
- [C3] Bathgate, K., **Bagchi, D.**, and Boyles S. D (2024). Data-Driven Modeling for Multimodal Port Resilience Assessment. *Presented in UT Austin Center for Transportation Research Annual Symposium, 2024*, Austin, USA.
- [C2] **Bagchi, D.**, Agarwal, P., Rambha, T., and Pandey, V. (2023). A Local Search Heuristic for Bi-criterion Steiner Traveling Salesman Problem. *Transportation Research Board 102nd Annual Meeting 2023 (TRB)*, Washington, D.C., USA.
- [C1] **Bagchi, D.**, Agarwal, P., Rambha, T., and Pandey, V. (2022). A Local Search Heuristic for Bi-criterion Steiner Traveling Salesman Problem. *Institute for Operations Research and the Management Sciences Annual Meeting 2022 (INFORMS)*, Indianapolis, USA.

OPEN SOURCE PROJECTS

Logit-SUE [\[Code\]](#)

Developer

- Among the fastest open-source Python solvers (as of 2026) for path-based stochastic user equilibrium traffic assignment under the Logit route-choice model.
- Implements Newton-GMRES with superlinear convergence, Barzilai-Borwein rules, and MSA variants with adaptive constant step sizes (linear convergence).
- Benchmarked on large-scale networks from the Transportation Network Test Problems, including Chicago Regional and Philadelphia.

Multimodal Analysis Port Simulation (MAPS) [\[Code\]](#) [\[Technical manual\]](#)

Co-developer

- Modular discrete-event simulation (DES) framework in Python for multimodal port operations across container, liquid bulk, and dry bulk terminals.
- Supports planning, bottleneck detection, and what-if scenario analysis using real-world data inputs, with sample port data included.
- Models capacity and resilience under operational scenarios such as channel expansion, hurricanes, and fog.

Last-Mile EV Logistics [Code]

Co-developer

- Codebase for safe and energy-efficient routing of last-mile electric freight vehicles, tested on Amazon delivery routes in Austin, TX (data included).
- Solves the bi-criterion Steiner Traveling Salesperson Problem with turn penalties using state-of-the-art heuristics.
- Also includes exact approaches via a mixed-integer linear program solved with CPLEX.

TEST SCORES

- Graduate Record Examination (GRE) – 155/170 (Verbal Reasoning), 167/170 (Quantitative Reasoning), 4.0/6.0 (Analytical Writing), 2023.
- Test of English as a Foreign Language (TOEFL) iBT Test – 29/30 (Reading), 29/30 (Listening), 26/30 (Writing), 25/30 (Speaking), 109/120 (Overall), 2023.
- All India Rank 40 in National Creativity Aptitude Test (NCAT), 2020.
- **99.3 percentile score in Joint Entrance Examination (Main)** out of 1.5 million candidates, 2019.
- **Among top 1% in Joint Entrance Examination (Advanced)** out of 120 thousand candidates, 2019.
- 96% in All India Senior School Certificate Examination (12th Board examination), 2019.
- CGPA 10/10 in All India Secondary School Examination (10th board examination), 2017.

TEACHING EXPERIENCE

- **Teaching Assistant**, CE 311S Probability and Statistics for Civil Engineers, University of Texas at Austin (*Spring 2025; Consolidated student feedback rating: 4.02 / 5.00.*)
 - Assisted with a major course redesign.
 - Led two weekly discussion and recitation sections of 20 students each.
 - Prepared assignment solutions and assisted with grading.
- **Teaching Assistant**, CE 311S Probability and Statistics for Civil Engineers, University of Texas at Austin (*Fall 2025; Consolidated student feedback rating: 4.22 / 5.00.*)
 - Led one weekly discussion and recitation section of 20 students.
 - Prepared assignment solutions and assisted with grading.
- **Engineering Teaching Assistant Certificate**, Cockrell School of Engineering, University of Texas at Austin (2025).

SKILLS AND INTERESTS

- **Research interests:** Freight and maritime logistics, traffic assignment, congestion pricing.
- **Technical skills:** Optimization, discrete event simulation, machine learning, data analysis, mathematical programming, Markov decision processes, reinforcement learning.
- **Programming languages:** Python (primary), C, Julia, GAUSS, SQL.
- **Coursework:**
 - **Transportation:** Transportation network analysis, dynamic traffic assignment, railway project design & construction, public transportation, traffic engineering, discrete choice modeling, infrastructure systems.
 - **Operations research and mathematics:** Linear programming, non-linear programming, optimization, applied stochastic processes, Markov decision processes, game theory, introduction to computing for AI & ML, linear algebra, real analysis.

REFeree SERVICE

Journal Referee

- Transportation Research Part B (2 reviews).
- Transportation Research Record (1 review).

Conference Referee

- Transportation Research Board Annual Meeting (8 reviews).

SOCIETY SERVICE

- **Community Service Chair**, UT Student Transportation Council, UT Austin chapter of Institute of Transportation Engineers (2023–2025)
- **Member**, Notebook Drive, Indian Institute of Science (2019–2023)

Co-CURRICULAR ACTIVITIES

- **Spanish Guitar**: Diploma with first-division distinction from Nikhil Bharat Sangeet Samiti.
- **Chess**: Member of All India Chess Federation, peak stable Lichess ratings ([@debojjalb](#)): 1423 (Rapid), 1453 (Blitz), 1175 (Bullet), 1502 (Puzzles).

LEADERSHIP ROLES

- Co-founded [CoachIO](#), an ed-tech startup to provide affordable boot-camp courses to KVPY and Olympiad aspirants across India. Managed a team of 9 members with gross revenue over 5 lakh INR.
- Coordinated and sourced funding for [Quadspark](#), a national level quiz competition as a part of Pravega, IISc, Bengaluru. The event witnessed 1200+ participants and was held in 3 stages.
- Content creator on personal YouTube channel [Debojjal Bagchi](#). The channel currently has over 10k subscribers and 1M+ views.